Java exam question ideas

Introduction to Java – JVM - Data Types, Variables, Operators, Expressions – Control flow Statements – Methods - Arrays – Classes and Objects –Constructors - Access Specifiers –Static members- this –constants- - String Class- Working with Data and Time API
PRACTICALS:
1. Develop programs using Java Basic Constructs and Arrays using any standard IDE like NETBEANS / ECLIPSE
2. Programs to illustrate concept of class and static classes and methods
3. Implement programs using string class, Date and Time API
UNIT – II POLYMORPHISM AND INHERITANCE 9L, 12P
Overloading Methods – Static, Nested and Inner Classes. Inheritance– super Classes and sub classes -Overriding –Downcasting – Operator instance of - Abstract Classes – final classes - Packages – Interfaces.
PRACTICALS:
1. Implement abstract classes, overloading and overriding
2. Implement Interfaces
UNIT – III EXCEPTION HANDLING AND MULTITHREADING 8L, 12P
Exception Handling – Java's Built-in Exceptions – User defined Exception - Assertions. Multithreading- Priorities–Synchronization – Avoiding Deadlocks– Wrappers– Autoboxing and Unboxing.
PRACTICALS:
1. Programs using Exception handling
2. Programs using Multi-Threading and synchronisation
UNIT – IV FILE STREAMS AND DATABASE 9L, 12P
Java I/O- Reading and Writing Files - Regular Expressions – Streams API - Object Serialization – Generic collections – Generic Classes - Lists, set, Maps – GenericMethods - Lambda expressions – Databases with JDBC.
PRACTICALS:
1. Programs using Streams API and File I/O for reading and writing the contents in sequential and random order.
2. Programs using Generics classes and methods
3. Event-driven programs for GUI applications with Database Connectivity
UNIT – V WEB DEVELOPMENT AND FRAMEWORKS 10L, 12P
Event handling: Event, Listeners and adapter classes, anonymous inner classes. Abstract Windowing Toolkit (AWT): Button, Label, Checkbox, Checkbox Group, Text Field, Text Area, Choice, List, Menu, Panel, Scrollbar and Swing components, Layout managers, Complex Components - Java Servlets - Apache Tomcat –Java Web Framework -Spring Boot
PRACTICALS:
1. Creation of servlet based applications
2. Design a Java Web application using Spring Boot

i have semester exam tomorrow
i am in Anna University
refer ceg and mit questiion paper 2023 revised syllsbus.
all the questions are seanariyo based question

list the important and twisting 8marks and 2marks quetions..

ex:

Second Assessment CS23304-Java - B.E (CSE) –II semester
Common Programming
to all Batches
Part-A (5X 22=10 marks)
Max.Marks: 50
2. Differentiate Explain how to print all the elements in a collection without using an Iterator.
between collections and streams 3. Write the lifecycle of Servlets
4. Compare and contrast between notify and notifyAll 5. Give a regular expression to generate an email id with the following format: first four
format:mm-dd-yyyy
letters of first name followed by first 5 letters of second name and date of birth in this @ test.com
6.
Part-B (10 X 3 = 30 marks) (Answer an THREE) Create a City class with name (string), population (double) and stateCapital(Boolean) indicating whether the city is a stateCapital or not and onTheSea (Boolean) - is it seashore city or no and include the necessary methods. Write a test class to print the following using Lambda expressions:
a. Print the cities that are on seashore as well as state capital
b. Print the cities sorted in alphabetical order and sorted in population
c. Print the cities that are stateCapital sorted by population d. Print the cities that are on the sea shore and sort alphabetically
7. An astronomical observatory has 3 telescopes available for public viewing. Multiple participants arrive at different times and want to view celestial objects through these telescopes. Each telescope can be used by only ONE participant at a time. Write a multithreading java program such that the Participants should be served synchronously.
8. A custom window shade designer charges a base fee of Rs.50 per shade. In addition, charges are added for certain styles, sizes, and colors as follows:
Styles Sizes: Colors
Regular shades: Rs.0 25 inches wide: Rs.0 Natural: Rs.5
Folding shades: Rs.10 27 inches wide: Rs.2 Blue: Rs.10.
Roman shades: Rs.15 32 inches wide: Rs.4 Teal: Rs.15
40 inches wide: Rs.6 Red: Rs.20.
Green: Rs.25
Create an application that allows the user to select the style, size, color, and number of shades from lists or combo boxes. The total charges should be displayed.
9. Write a servlet program to receive the username and password and display the same.Part-C (1X 10 = 10 marks)
Develop
manage
a
bookings.
Truck Booking Management System using HashMap data structures to eficiently
class Truck {
private String registrationNumber; // e.g., "TN01AB1234"
private String truckType; // "SMALL", "MEDIUM","LARGE","CONTAINER"
private double capacityTons; // Capacity in tons

private boolean isAvailable; // Current availability status
// Constructor and methods
class Booking {
private String bookingId; // e.g., "BOOK001"
private String truckRegNo;
private String route; // e.g., "Chennai-Bangalore"
private double distance; // in kilometers
private String bookingDate; // Format: "YYYY-MM-DD"
private double totalCost;
// Constructor and methods
class TruckBookingSystem {
private HashMap<String, Truck> truckFleet; // RegNo -> Truck
private HashMap<String, Booking> bookings; // BookingId -> Booking
private HashMap<String, Double> routeDistances; // Route -> Distance
// Constructor public TruckBookingSystem(){
// Initialize all HashMaps
}
}
Implement the following functions
Add a new truck to the fleet - return true if added successfully, false
if truck already exists
public boolean addTruck (Truck truck) {
// Your implementation (2 marks) }
Add a predefined route with distance public void addRoute(String route, double distance) //
Your implementation (2 marks)
available Book a truck - return Booking object if successful, null if truck not
public Booking bookTruck(String truckRegNo, String route, String bookingDate)First
Assessment – B.E (CSE) - III semester
CS23304- Java Programming
Common to all Branches
Part-A (5X 2 = 10 marks)
Max.Marks: 50
1. Once an array is created, its size cannot be changed. Does the following code resize the
array?
int[] num;
2.
num = new int[10];
num = new int[20];
Identify and correct the errors in the following program:
public class Test {
public static method1(int n, m) {
n += m;
method2(3.4);
}
}
public static int method2(int n) {
if (n > 0) return 1;
else if (n == 0) return 0;
else if (n< 0) return -1;
3. Suppose that s1, s2, s3, and s4 are four strings, given as follows:
String s1 = "Welcome to Java";

String s2 = s1;
String s3 = new String("Welcome to Java");
String s4 = "Welcome to Java";
What will be the output for the code snippet?
s2.equals(s3) s1.compareTo(s2)
4. Compare and contrast overriding and overloading.
5. List the advantages of using exception handling. 9x10 =30
6.
Part-B (X=30 marks)
Answer any THREE
Create a class Book that will record the books purchased by a person. It will have the attributes
• booksPrices-an array of double values that are the prices of all books
booksPurchased-the number of books purchased so far
maxNumberBP-the maximum number of book purchases that can be recorded
and the following methods:
• Books(max)-a constructor that sets the maximum number of book purchases to max
• addPrice(p)-adds to an array a price whose value is p
• getNumberOfBooksPurchased-returns the number of books purchased
getTotalBooksPrice-returns the total price of the books purchased
• getAverageBookPrice()-returns the average price of all the books purchased
• getCountAbove(p)-returns the number of books that are greater than p in value
Create a driver class to create object of type Book and test all Create an interface
MessageEncoder that has a single abstract method String encode(plainText), where
plainText is the message to be encoded and returns the encoded message. Create a class
SubstitutionCipher that implements the interface MessageEncoder. The class variable is
shift(int)
initialized by constructor that takes one parameter called shift. Override the method encode
such that
each letter is shifted by the value in shift. For example, if shift is 3, a will be replaced by d, b will
be replaced
by e, etc.
Create a class ShuffleCipher that implements the interface MessageEncoder with an attribute
shuffkenumber initialized by the one argument constructor.Overrde the method encode so that
the
message is shuffled n times. To perform one shuffle, split the message in half and then take
characters
from each half alternately. For example, if the message is abcdefghi, the halves are abcde and
fghi. The
shuffled message is afbgchdie.
Suppose songs is the input String, write a program to prompt the user for a song number and
then
extract the song number plus the rest of that string's line from a given input string songs and
print it
using class String or String builder methods
String songs =
"1. Bow Wow - Fresh Azimiz\n" +
"2. Weezer - Beverly Hills\n" +
"3. Dave Matthews Band - Crash Into Me\n" +
"4. Sheryl Crow - Leaving Las Vegas\n";
9. Implement the following exceptions for the Question no.6:
i. The Books(max) constructor throws IllegalArgumentException if max is initialised with zero

and
booksPurchased is initialised with zero
ji When booksPrices is accessed outofbounds throw ArrayIndexOutOfBoundsException
iii. The driver class throws InputMismatchException when the input obtained from user
iv.
V.
mismatches with the required type
addPrice(p) throws Custom Exception whenever the value of p is negative or zero
Include finally block wherever neccessary
Part-C (10 marks)
10. Create an abstract class called Account that has datamembers: accno(int), accname(String) and
balance(double), contactAddress(Address). The member methods are: parameterized constructor and
toString method. abstract methods: void deposit(double amt) and double withdraw(double amt)
Derive two classes from class Account: SavingsAccount and CheckingsAccount. The members of
SavingsAccount are noofTransactions(int), parameterized constructor, overridden methods toString(),
deposit(double amt) that increments balance with amt, withdraw(double amt) that decrements the
balance by amt and noofTransactions is incremented by 1 inside deposit() and withdraw().The members
of CheckingAccount are parameterized constructor, overridden methods deposit(double amt) that
increments balance with amt, withdraw(double amt) that decrements the balance by amt only if the
balance is above 1000 after decrementation and toString().
Define TestAccount that instantiates objects of SavingsAccount and CheckingsAccount. The methods of
the derived class are tested. Check the SavingsAccount object is an instance of class Account.

Here are **some likely 8-mark and 2-mark scenario-based question ideas** for your upcoming exam **Anna University (CEG/MIT) – CS23304 Java Programming (Revised Syllabus 2023)**. These are derived by mapping the syllabus units you shared to typical question styles. **Use these as practice drills**, not guaranteed exact questions.

✅ 2-Mark question ideas

These are short, direct, often definition/comparison style, or very small scenario based.

1. What is the role of the Java Virtual Machine (JVM) in Java runtime?
2. Define what a “primitive data type” is in Java and give two examples.
3. What is the difference between `==` and `equals()` when comparing objects in Java?
4. What is a constructor? When is it called?
5. What is the meaning of `static` in Java? Give one example use.
6. What is an inner class?

7. What is multithreading? Name one benefit of using threads.
8. What is the difference between checked and unchecked exceptions in Java?
9. What is a generic class? Give a simple example (just signature).
10. What is the difference between `HashSet` and `TreeSet`?
11. What is JDBC used for in Java?
12. What is the lifecycle method of a Servlet when it is first loaded by the container?
13. What is a lambda expression in Java? Give one simple example.
14. What is the purpose of `synchronized` keyword in Java threads?
15. What is the role of `final` keyword when applied to a class?

⚠️ 8-Mark question ideas

These will be scenario-based, requiring you to write code, explain workings, compare alternatives, or design small systems. They will map to the units in your syllabus.

Unit I: Basics, Classes & Objects

1. **Scenario**: "Write a Java program to manage student records. Each student has name, roll-no, and date of birth. Use arrays to store multiple students. Provide methods to add a student, list all students, and display students born after a certain date. Also demonstrate using the String class and the Date/Time API to parse and format the date of birth."
2. Explain the difference between `this` keyword, instance variables and local variables with a practical example. What happens if you don't use `this` when referencing instance variables?
3. Write a class `BankAccount` with attributes accountNumber, accountHolderName, balance. Provide constructors, methods `deposit`, `withdraw`, and override `toString()`. Demonstrate use of static member to count number of accounts created.

Unit II: Inheritance & Polymorphism

4. Scenario: "You are building a transport system. Create an abstract class `Vehicle` with methods `start()` and `stop()`. Then extend `Car`, `Bus` subclasses. Demonstrate method overriding, downcasting, use of `instanceof`, and also one nested/inner class inside `Vehicle` which handles maintenance schedule."
5. Explain method overloading vs method overriding. Provide code examples showing where each one is used. Also, show how you can prevent method overriding using a `final` class or method.
6. Design an interface `Payment` with method `pay(double amount)`. Create two implementing classes `CreditCardPayment` and `UPIPayment`. Also, show use of packages and how you import classes from other package.

Unit III: Exception Handling & Multithreading

7. Scenario: "Write a Java program where multiple threads simulate customers using ATMs. Each ATM object can only serve one customer thread at a time. Use synchronized blocks and priorities. Also

handle possible exceptions like **`InsufficientFundsException`** (user-defined) and demonstrate use of assertions."

8. Explain how synchronization prevents race conditions in Java threads. What is a deadlock? Show, by code snippet, how a deadlock might occur and how you would prevent it.
9. Write a try/catch/finally block where you open a file, read data, handle **`IOException`**, and then close the resource in **`finally`**. Also explain autoboxing/unboxing with wrappers in Java (e.g., **`Integer`** vs **`int`**).

Unit IV: File Streams, Collections, Streams API & Databases

10. Scenario: "You are writing a Java program for a small library. Use **`List<Book>`** to store books, **`Map<String,Integer>`** to map book titles to quantities. Use generics and generic methods to find all books below a given price. Then write code to save the list to a file using object serialization. Later read it back. Also write code that uses a database (via JDBC) to store the same information."
11. Write a program using Java Streams API to process a list of employees and print the names of employees whose salary > X sorted by age. Also explain difference between collections and streams.
12. Explain regular expressions in Java. Provide a regex pattern to validate an email id of the format: first four letters of first name + first five letters of second name + birthdate (mm-dd-yyyy) @ test.com. Then write code snippet that uses **`Pattern`** and **`Matcher`**.

Unit V: Web Development & Frameworks

13. Scenario: "Write a servlet mapped to **`/login`** that receives **`username`** and **`password`** as HTTP POST parameters. In **`doPost`**, validate that both parameters are non-empty. If OK, forward to welcome page showing the user name. Otherwise redirect to an error page. Explain the servlet lifecycle methods and how the servlet container (e.g., Apache Tomcat) manages them."
14. Use Spring Boot to create a simple REST endpoint **`/books`** that returns a list of books in JSON. Also explain how you would connect to a database and use JPA to retrieve books.
15. Describe event handling in Java GUI (AWT/Swing). Design a GUI window with a button, text field and label. On button click, text from text-field is shown in label. Also explain layout managers and one example of using anonymous inner class as event listener.

🔍 Tips for Preparation

- Focus on **scenario-based coding questions**: you'll likely be asked to *write* a small program or class design rather than just definitions.
- Practice writing **complete code snippets**: class definitions, methods, interface implementations.
- Be comfortable with **generics, Streams API, lambdas**, as they are newer topics and often used in scenario questions.
- Understand **threading concepts** with synchronization and deadlock, and be able to write simple multi-thread code.
- For web part: know servlet lifecycle (**`init()`**, **`service()`**, **`destroy()`**), doPost/doGet methods, mapping annotations or **`web.xml`**, basic Spring Boot setup.

- Review previous year question papers from Anna University (CEG/MIT) since the theme "scenario based" and part-A/part-B/part-C structure is consistent.
- Time yourself: For 8-mark questions aim ~15-20 minutes each; for 2-mark questions ~2-3 minutes each.
- Use pen and paper (or IDE) to write code but also **explain** your design in words, especially for part-B/part-C style questions.

If you like, I can **fetch actual last 3 years question papers** for CS23304 (CEG/MIT) and **highlight the most repeated twist questions** for you. Would you like me to do that?